

Product Carbon Footprint Study Report

Product Name: inverter, 2.5kW

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: inverter, 2.5kW

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item inverter, 2.5kW

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item inverter, 2.5kW.

2. System Boundary

inverter production, 2.5kW, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of inverter, 2.5kW produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **223.211 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

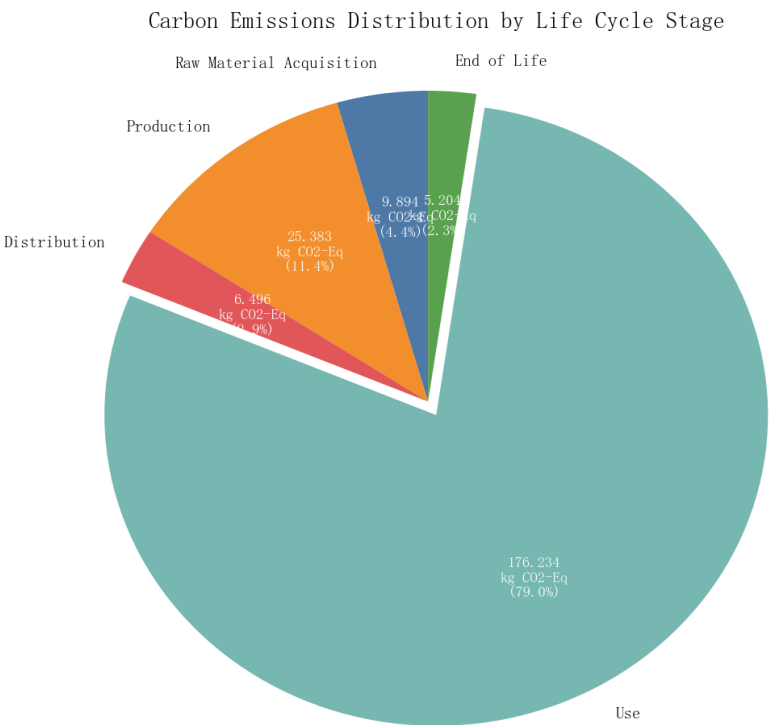


Table 1: Carbon Emissions by Life Cycle Stage

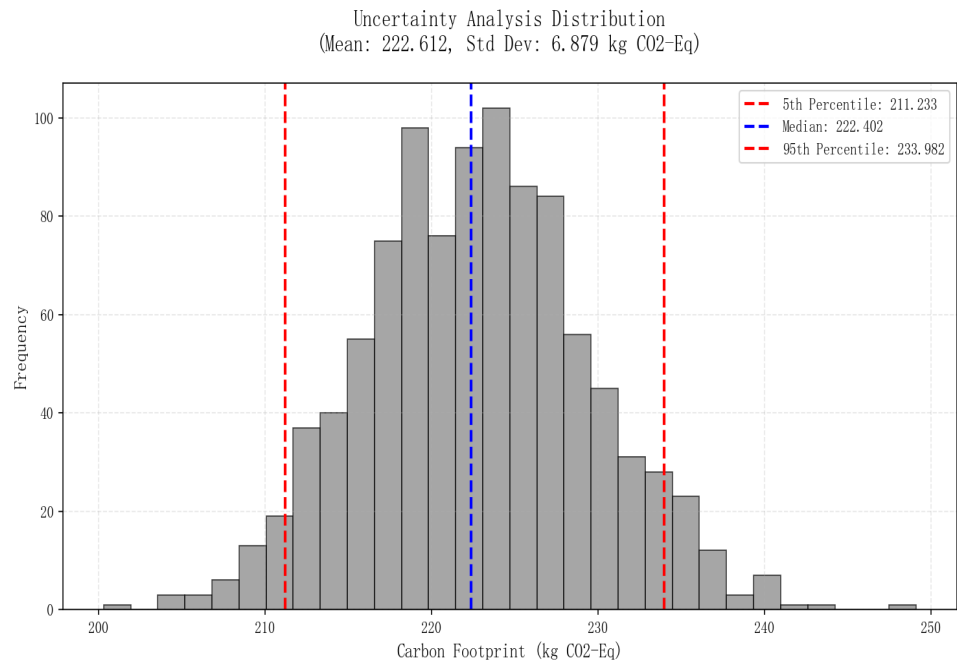
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	9.894	4.43
Production	25.383	11.37
Distribution	6.496	2.91

Use	176.234	78.95
End of Life	5.204	2.33
Total	223.211	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	222.612
Standard Deviation	6.879
Coefficient of Variation (CV)	3.09%
Median	222.402
5th Percentile	211.233
95th Percentile	233.982

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the inverter (2.5 kW) is estimated at 223.211 kg CO₂-eq per declared functional unit. This is a relatively moderate impact for an electronic power-conversion device, but it is strongly dominated by the use phase. The use stage contributes 176.234 kg CO₂-eq, representing 79.0% of the total footprint. This clearly indicates that the inverter's operational energy consumption—and the related electricity mix—is the most critical factor controlling life cycle emissions.

The remaining life cycle stages are comparatively minor: production contributes 25.383 kg CO₂-eq (11.4%), raw material supply contributes 9.894 kg CO₂-eq (4.4%), distribution contributes 6.496 kg CO₂-eq (2.9%), and end-of-life treatment contributes 5.204 kg CO₂-eq (2.3%). From a product design perspective, this is a typical profile for energy-using equipment with a long operational lifetime: the greatest reduction potential lies not in materials or manufacturing, but in improving energy efficiency and reducing electricity losses during use.

The largest emission source can therefore be clearly attributed to the use phase, most plausibly from electricity consumption associated with inverter power losses, standby operation, and load-dependent conversion inefficiency. No process-level contributions could be reported because the top process entries were marked as “Unknown” with zero values; this represents a data limitation rather than an indication of zero impact.

2. Statement of Assumptions and Limitations

Several key assumptions were made in this study. The use-stage emissions are assumed to be calculated from a defined inverter load profile, an assumed service lifetime, and a specific electricity grid mix. These parameters strongly influence the result, as the use phase accounts for nearly four-fifths of the total carbon footprint. The production and raw material stages are likely based on generic datasets for electronic components, printed circuit boards, metals, and housings rather than supplier-specific inventory data.

A major limitation is that the top process contribution table contains only “Unknown” entries with zero values. This suggests that process-level allocation and mapping were incomplete or that the LCA software could not reconcile certain inventory flows with recognized impact-category processes. As a result, the contribution analysis at the individual process level cannot be interpreted with confidence. The uncertainty analysis provides some reassurance: the Monte Carlo simulation with 1,000 iterations gave a mean of 222.612 kg CO₂-eq, a coefficient of variation of only 3.09%, and a 90% confidence interval of 211.233–233.982 kg CO₂-eq, indicating that the total result is statistically stable under the current model assumptions.

The system boundary is cradle-to-grave, covering raw material supply, production, distribution, use, and end-of-life treatment. However, capital equipment, infrastructure, and auxiliary consumption not directly related to the inverter's operation are likely excluded. The study should be interpreted within these boundaries, and direct comparison with other products is only valid if identical assumptions and functional units are used.

3. Suggestions for Improvement

- Prioritize use-phase efficiency improvements: reduce standby power consumption, improve power conversion efficiency, and implement intelligent standby/sleep modes. Since use accounts for 79% of the total footprint, even a 5% cut in operational energy losses would save roughly 8.8 kg CO₂-eq per unit.
- Promote the use of renewable electricity during the operational phase: publish product information encouraging customers to pair the inverter with on-site solar generation or green electricity tariffs, effectively reducing the grid emission factor applied in the use stage.
- Substitute lower-carbon materials and increase recycled content in the housing, heat sinks, and conductive components. This can directly lower the raw material stage contribution (currently 4.4%) and reduce embodied emissions in aluminium, copper, and plastics.
- Collect supplier-specific primary data for the main components, especially power semiconductors, control boards, and casings, to replace generic “Unknown” process entries. This will improve process-level transparency and enable targeted hot-spot reduction.
- Disaggregate the use-stage model by actual inverter loading profiles and regional electricity mixes. This will reduce uncertainty and provide more credible, market-specific carbon footprint results for stakeholders.